Einstein and Machine Learning

The Photoelectric Effect as Statistical Inference

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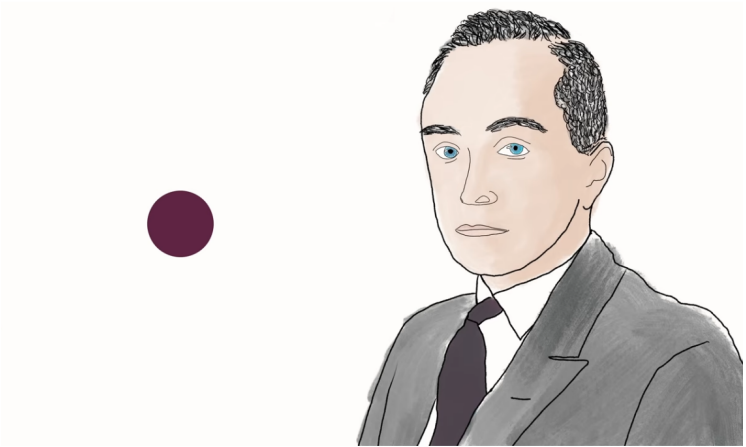
The Born Rule



Weirdest Rule in Physics

Claim

When Einstein proposed the light-quantum hypothesis in 1905 to explain the photoelectric effect, he was doing **machine learning** and **statistical inference**.

1. **Theory** (1905): Einstein proposes a manifold of statistical models (an "hypothesis class")
2. **Data** (1887–1916): experimenters collect data
3. **Estimation** (1916): Millikan runs Einstein's regression and recovers Planck's constant to 0.5%

The Photoelectric Effect and Feature Engineering

The phenomenon (Hertz, 1887): shine light on a metal; electrons are ejected. Apply a stopping potential V_0 to halt the fastest electrons.

Lenard (1902) ran an ablation experiment—measured V_0 as a function of both frequency ν and intensity I:

Feature	Effect on V_0	Importance
Frequency ν	Strong $\uparrow$	**High**
Intensity I	None	**Zero**

Classical wave theory predicted the opposite: V_0 should rise with I and be independent of ν.

ML translation: feature-importance analysis eliminates I; selects ν as sole relevant input.

Einstein's Model and Inductive Bias

A light quantum carries energy $E = h\nu$. A single quantum is absorbed by a single electron. Energy conservation gives:

$$\boxed{V_0 = \frac{h}{e}\nu - \frac{W}{e}}$$

- Slope $= h/e$ is **universal** across all metals
- Proposed before reliable V_0-vs-ν measurements existed

The quantum hypothesis restricts the hypothesis class from *all functions* of $(\nu, I, \lambda, \ldots)$ to an affine function in ν alone with a universal slope.

Hypothesis class	Dimension	Data needed
All measurable functions of $(\nu, I, \ldots)$	Infinite	$\gg 10^6$
Affine in ν alone (Einstein)	2	~ 5

The Regression Problem

Adding noise and a nuisance parameter:

$$V_{0i} = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\beta}_{h/e} \nu_i + \underbrace{\varepsilon_i}_{\text{noise}}, \qquad \alpha = -\frac{W}{e} + \underbrace{v_c}_{\text{contact EMF}}$$

- **Training data:** $\{(\nu_i, V_{0i})\}$
- **Task:** estimate $\beta = h/e \Rightarrow$ Planck's h
- **Loss:** squared error (OLS = MLE under Gaussian errors)
- **Nuisance:** v_c can be as large as 3 V; shifts intercept but **does not bias slope** if constant within a run

Computing the regression is trivial once you have data. The 29-year gap between Einstein and Millikan is a data engineering problem, not a statistical one.

29 Years of Collecting Data (1887–1916)

Experimenter	Year	Data quality	Problem
Hertz / Hallwachs	1887–88	Qualitative only	No labeled (ν, V_0) pairs
Lenard	1902	Qualitative	No reliable numerical V_0
Hughes	1912	3 UV lines	Contact EMF uncontrolled; 65% scatter in h
Richardson & Compton	1912	8 metals	Same problems; 60% scatter
Millikan	**1916**	**Clean surfaces, wide $\Delta\nu$, controlled v_c**	**h to 0.5%**

Millikan's Experimental Design and the Cramér–Rao Bound

Key data-quality innovations ("a machine shop in vacuo"):

- **Fresh surfaces:** electromagnetic knife shaved alkali metal in vacuum—no oxidation, no work-function contamination
- **Spectrally pure lines:** mercury arc with filters to exclude UV companions
- **Wide frequency range:** 5461 Å to 2535 Å ($\Delta\nu = 6.3 \times 10^{14}$ Hz, a factor of two)
- **Controlled nuisance:** Kelvin double-balance measured v_c; stable to ± 0.01 V over 9 hours

Cramér–Rao bound:

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

Dataset	n	$S_{\nu\nu}$ (10^{28} Hz^2)	SE($\hat{\beta}$) (10^{-18} V·s)
Restricted range (like Hughes)	3	1.96	10.0
Millikan full 6-pt	6	28.14	2.6

Millikan's Data and OLS Result

λ (Å)	ν (10^{14} Hz)	V_0^{Na}	V_0^{Li}
5461.0	5.490	0.454	—
4339.4	6.908	1.039	0.499
4046.8	7.409	1.246	0.706
3650.2	8.213	1.578	1.037
3125.5	9.591	2.147	1.606
2534.7	11.827	3.030	2.529

OLS on sodium (5 points):

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = 4.129 \times 10^{-15}\ \mathrm{V{\cdot}s}$$

R^2	≈ 0.99999
SE($\hat{\beta}$)	4.6×10^{-18} V·s
Implied h	6.573×10^{-34} J·s
Error vs. NIST	-0.80%

h to **0.5%** from **5 data points**.

Universality: Zero-Shot Transfer Across Metals

The quantum hypothesis predicts $\beta = h/e$ is the same for *all* metals.

Separate regressions:

$$\hat{\beta}_{\mathrm{Na}} = 4.129 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\hat{\beta}_{\mathrm{Li}} = 4.137 \times 10^{-15}\ \mathrm{V{\cdot}s}$$
$$\text{difference} = 0.19\%$$

Wald test of $H_0 : \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$: $z \approx 0.87$ ($\ll 1.96$). Cannot reject universality.

In ML language:

- **Pre-training:** fit slope on sodium
- **Zero-shot transfer:** apply slope to lithium
- **Fine-tuning:** only the intercept (work function)
- **Transfer error:** 0.2%

Historical Estimates of Planck's Constant

Authors	Year	h (10^{-27} erg·s)	Error
Hughes	1912	3.55–5.85	−12 to −46%
Richardson & Compton	1912	3.5–5.8	−12 to −47%
Planck (black body)	1913	6.415	−3.2%
Millikan (Na, 5-pt)	1916	6.573	−0.80%
Millikan (Li)	1916	6.584	−0.64%
Millikan mean	**1916**	**6.579**	**−0.72%**
Modern NIST 2022	2022	6.6261	0

Pre-Millikan estimates: biased downward 10–47%.
Three sources: stray UV, narrow $\Delta\nu$, uncontrolled v_c.
Millikan eliminated all three.

Millikan's Doubts

Millikan confirmed Einstein's equation to 0.5%. Then wrote:

> *"Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found* **so untenable** *that Einstein himself, I believe, no longer holds to it."*
>
> Millikan (1916), p. 388

He had hoped the experiment would **refute** the equation.

- Confirming a regression equation says little about the structure that generated it
- Compton scattering (1923) finally convinced him
- Millikan's 1923 Nobel: "...first direct experimental proof ...of the complete validity of the Einstein equation"

ML-to-Physics Dictionary

Scientific step	**ML name**	**Who**
Propose $E = h\nu$; derive $V_0 = (h/e)\nu - W/e$	Inductive bias + hypothesis class	Einstein (1905)
V_0 depends on ν not I	Feature engineering	Lenard (1902)
Measure (ν_i, V_{0i}) pairs	Labeled training data	Millikan (1916)
Control contact EMF v_c	Nuisance-variable identification	Millikan (1916)
Min. $\sum(V_{0i} - \hat{V}_{0i})^2$	MSE training (OLS/MLE)	Millikan (1916)
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Test accuracy	Millikan (1916)
Na and Li give same $\hat{\beta}$	Zero-shot transfer learning	Millikan (1916)

Einstein vs. a Deep Neural Network

Epoch	n	Einstein model	DNN (VC $\approx$ 5700)	DNN outcome
Millikan 1916	5	Sufficient*	Insuf. ($\times$1140)	Interpolates; no inference
Modern era	$\sim$ 5000	Sufficient	Sufficient	Recovers line; cannot identify h/e

*Functional form from theory; two free parameters remain.

A three-layer DNN (16 neurons/layer): $p \approx 593$ parameters, VC dim ≈ 5700. Reliable 1% generalisation requires $n \gtrsim 5700$.

Einstein's theory made five training points sufficient—not more data, not a better algorithm.

DNNs win for problems without a known parametric family (images, proteins, language). The photoelectric problem sits at the opposite extreme.

Unintended Consequence: Born's Rule

Einstein's quantum hypothesis required $|E|^2 \propto$ photon density. The classical wave intensity $|E|^2$ becomes a probability density. Einstein called this the *Gespensterfeld* (ghost field).

Max Born (1926) formalised the same logic for the Schrödinger wave function:

$$\boxed{|\psi|^2 = \text{probability density}}$$

Three Nobel Prizes:

- 1921: Einstein—for the equation that made the measurement possible
- 1923: Millikan—for the measurement that confirmed it
- 1954: Born—for the probabilistic interpretation the equation made *inevitable*

Einstein spent the rest of his life rejecting Born's rule. He won the Nobel for a paper whose logic seeded the interpretation he most opposed.

Summary

The Thesis

Einstein was doing machine learning in 1905—but the step no ML system has automated is **specifying the hypothesis class**.

1. **Theoretical priors guide design.** $E = h\nu$ fixes the functional form; 5 training points suffice for 0.1% precision.
2. **Data quality is the binding constraint.** The regression is trivial; assembling good data took 29 years.
3. **Confirming an equation $\neq$ endorsing the model.** Millikan confirmed the slope to 0.5% while rejecting the light-quantum interpretation.

A data-driven system given only (ν_i, V_{0i}) pairs can recover $\hat{\beta} = 4.129 \times 10^{-15}$ V·s but cannot know it is h/e. **Physical meaning requires physical theory.**

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